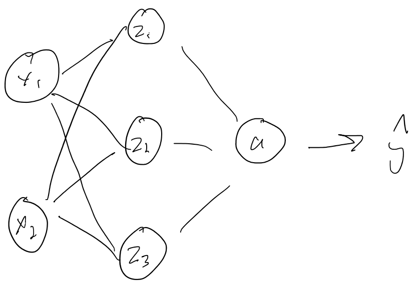
Algorithm practice

Friday, January 16, 2026 3:12 PM



$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$a = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$L = -\sum_i y_i \log(\hat{y}_i)$$

$z^{n \times n}$ = num_x · num_vocab

Vocab = ["eat", "sleep", "code"]

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad b = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$W = \begin{array}{cc} \begin{bmatrix} 0.5 & 0.5 \\ 1.0 & -1.0 \\ 2.0 & 0.0 \end{bmatrix} & \begin{matrix} z_1 \\ z_2 \\ z_3 \end{matrix} \\ \begin{matrix} x_1 & \quad x_2 \end{matrix} & \end{array}$$

True value = "code"

["eat", "sleep", "code"]
= [0 , 0 , 1]

$z_1 = 0.5(1) + 0.5(2) + 0$
$= 1.5$

$z_2 = 1.0(1) + -1.0(2) + 1$
$= 0$

$z_3 = 2.0(1) + 0.0(2) - 1$
$= 1.0$

$z = [1.5, 0, 1.0]$

$e^{z_i} = [4.48, 1.0, 2.72]$

$\sum e^{z_j} = 4.48 + 1.0 + 2.72$
$= 8.2$

$$\hat{y} = \left[\frac{4.48}{8.2}, \frac{1.0}{8.2}, \frac{2.72}{8.2} \right]$$

$= [0.546, 0.122, 0.332]$

$L_1 = -0(\log(0.546))$
$L_1 = -0$

$L_2 = -0(\log(0.122))$
$= 0$

$L_3 = -1(\log(0.332))$
$= 1.1026$

$L = [0, 0, 1.1026]$

Back propagation:

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = [0.546, 0.122, -0.668]$$

$$\frac{\partial L}{\partial w} = \partial z \cdot x^T = \begin{bmatrix} 0.546 \\ 0.122 \\ -0.668 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 0.546 & 1.092 \\ 0.122 & 0.244 \\ -0.668 & -1.336 \end{bmatrix}$$

$\eta = 0.1$

$W = W_{old}\ \eta\ \partial w$

$$= \begin{bmatrix} 0.5 & 0.5 \\ 1.0 & -1.0 \\ 2.0 & 0.0 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 0.546 & 1.092 \\ 0.122 & 0.244 \\ -0.668 & -1.336 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0.5 \\ 1.0 & -1.0 \\ 2.0 & 0.0 \end{bmatrix} - \begin{bmatrix} 0.055 & 0.109 \\ 0.012 & 0.024 \\ -0.067 & -0.134 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 - 0.55 & 0.5 - 0.109 \\ 1.0 - 0.012 & -1.0 - 0.24 \\ 2.0 - (-0.067) & 0.0 - (-0.134) \end{bmatrix}$$

$$= \begin{bmatrix} -0.05 & 0.391 \\ 0.988 & -1.24 \\ & 0.134 \end{bmatrix}$$

$$
= \begin{bmatrix} -0.05 & 0.391 \\ 0.988 & -1.24 \\ 2.067 & 0.134 \end{bmatrix}
$$